# Introduction to Blockchain and Smart Contracts

## Lecture 1: Cryptography

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# Nuts and Bolts — Cryptography

# What are the requirements?

- Every object of value must have certain invariants:
  - Scarce
  - Unique (must have measures in place to detect counterfeiting)
  - Immutable
- A possible solution in the digital realm is through cryptography
  - Cryptography: Provides a mathematically rigorous mechanism by which information can be made to have the following properties:
    - Confidential
    - Integrity
    - Non-repudiability

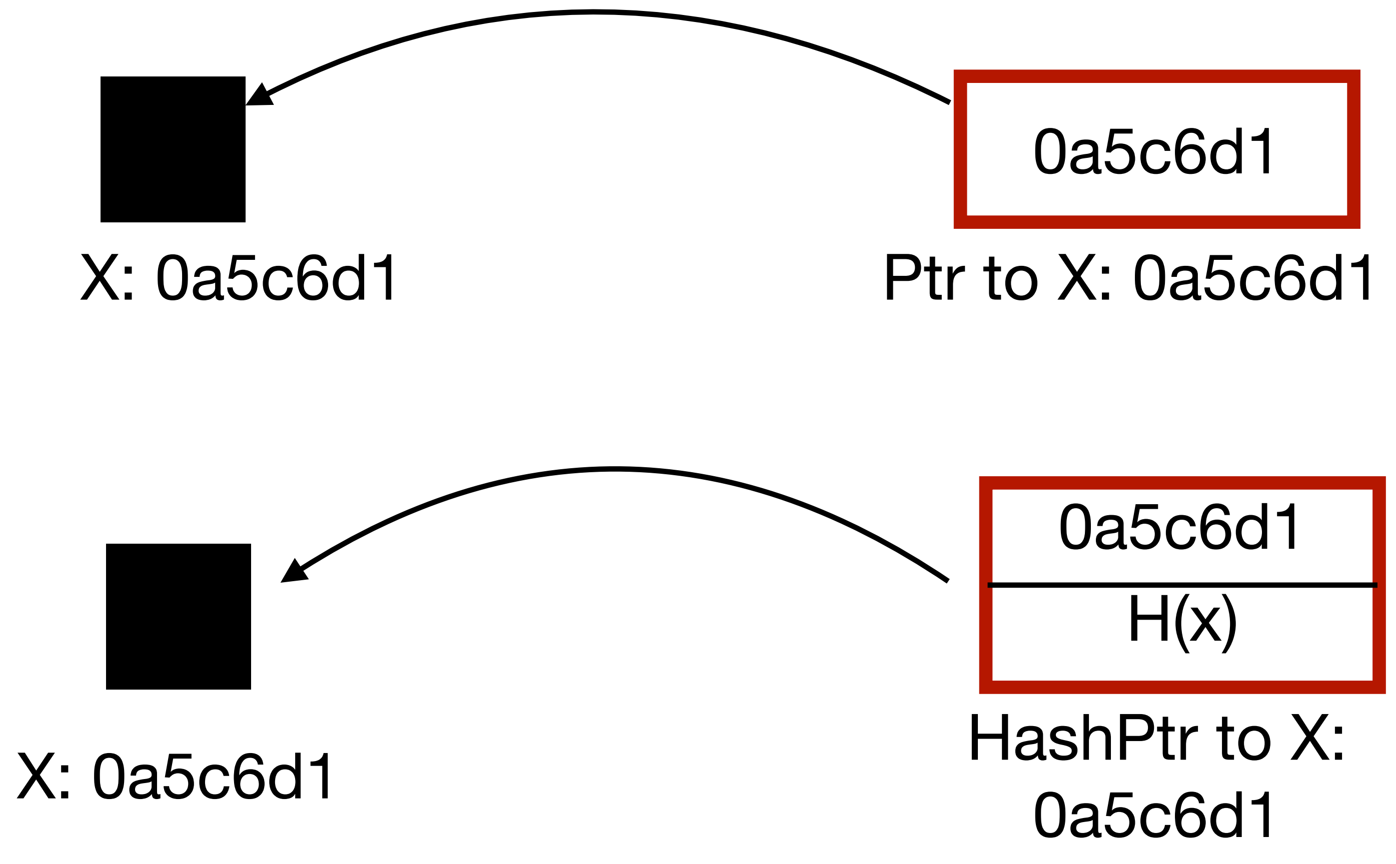
# Integrity: Cryptographic Hash Functions

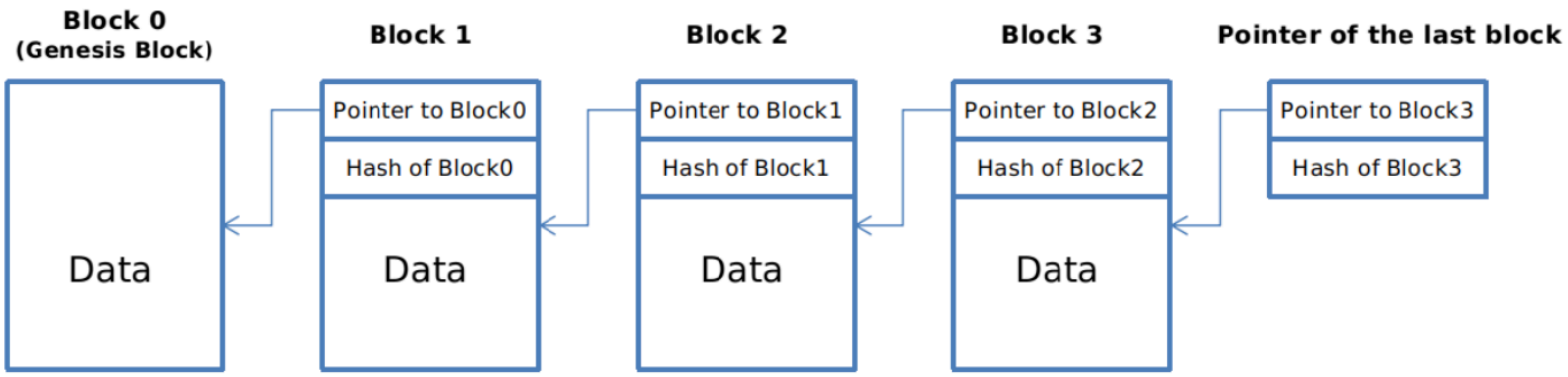
- A hash function $H(\,.\,)$ that has the following features:
  - Takes in arbitrary-sized input and produces a fixed-size output
  - It is a non-invertible function. That is, given an output one cannot compute the input
  - Forward computability is eminently efficient — typically O(n) where n is the input size
- For hash functions to be secure following invariants must be satisfied:
  - **Collision-resistance**: It is infeasible to find two values, $x, y$, s.t. $x \neq y$ yet $H(x) = H(y)$ with high probability
  - **Hiding**: Given $y$ s.t $y = H(x)$, there is no feasible way to find $x$
    - In fact it is $H(r\,||\,x)$ where $r$ is chosen from a distribution with high min-entropy
  - **Puzzle-friendliness**: if for every possible n-bit output value $y$, if $k$ is chosen from a distribution with high min-entropy, then it is infeasible to find $x$ such that $H(k\,||\,x) = y$ in time significantly less than $2^n$.

# From Hashes to Hash Pointers

## From Integrity to Tamper-evidence

- A pointer: object that stores mem address
- A hash pointer additionally stores cryptographic hash.
- Tamper-evidence property

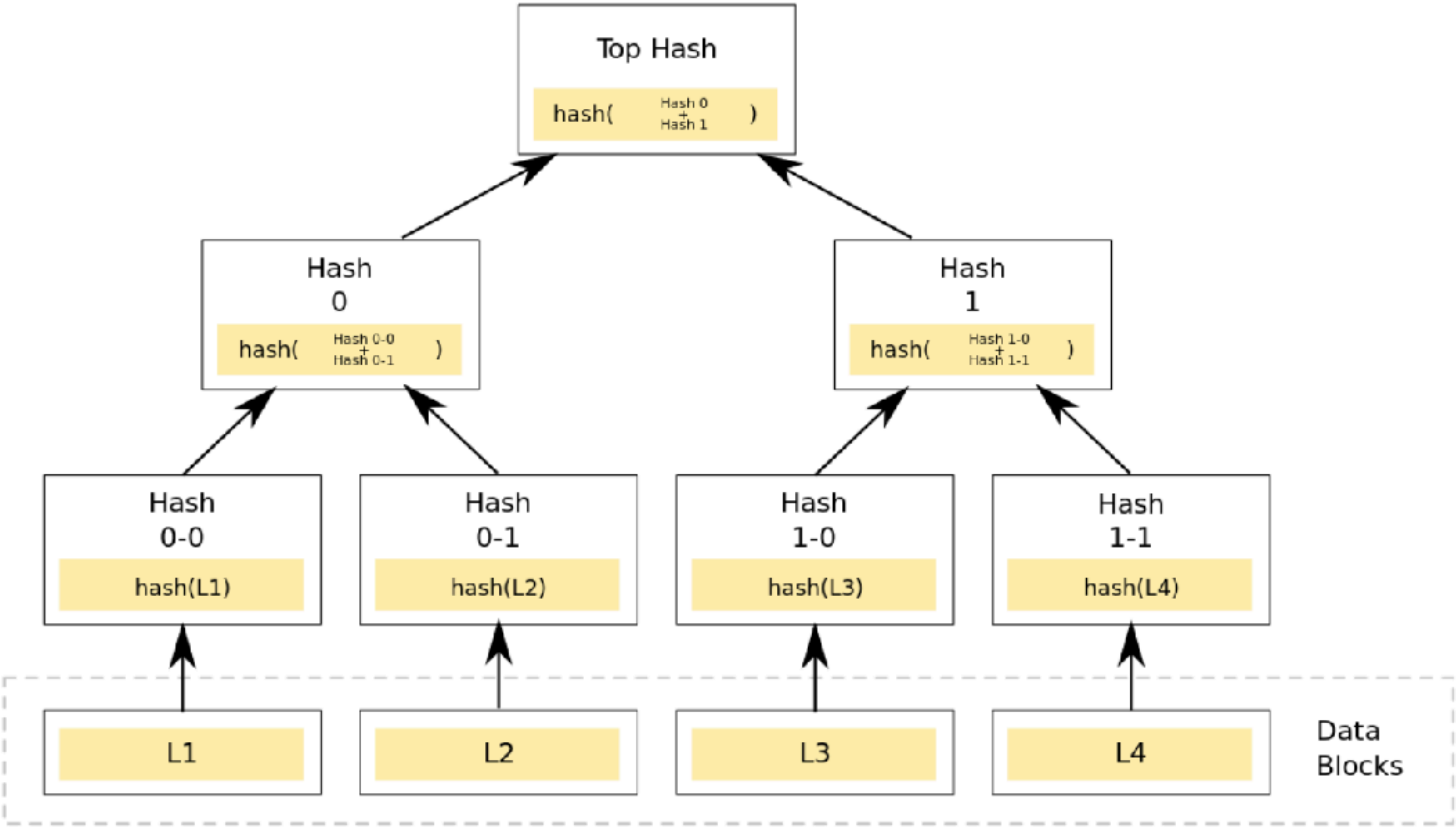


# Merkle Trees

## Moving towards efficiency

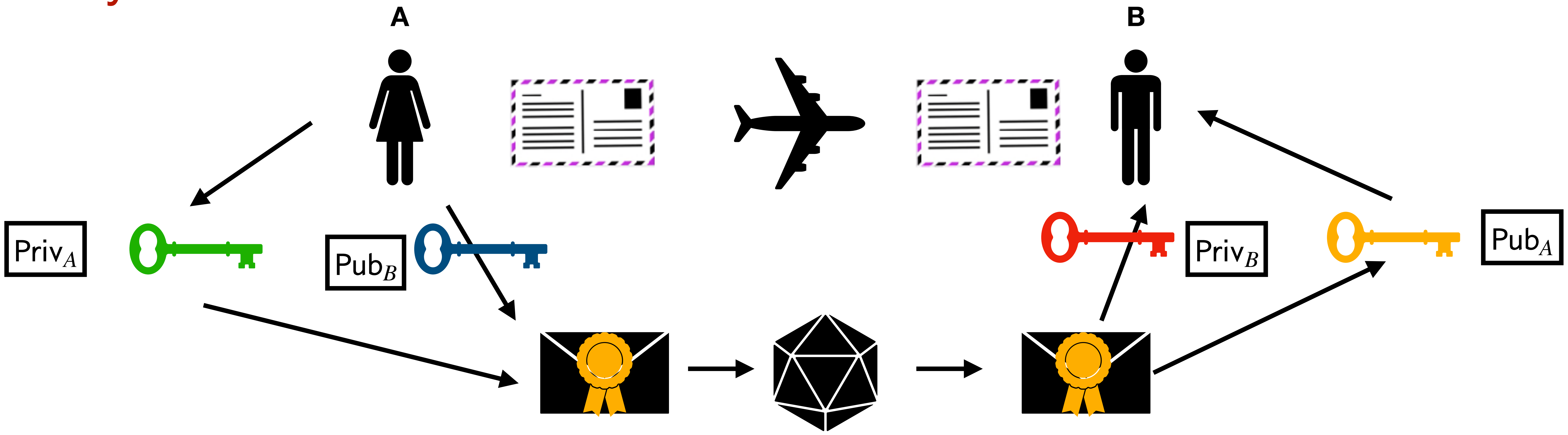
- Merkle tree is a binary tree of hash pointers
- Quick tamper-evidence check
- Concise proof of membership
- Proof of non-membership in log time and space



# Digital Signatures

## Identities and non-repudiation

- Properties:
  - Cannot be forged; Cannot be snipped off
- Public keys as identities — can make as many of them as one desires; need to be careful of sybil attacks



# Digital Signatures

## ECDSA common in Blockchain

- How are they different from MAC?
- Engineering Truth —> Hash then sign!
- ECDSA setting:
    - Given a cyclic group generator $G$ on an elliptic curve with curve order $n$
    - Private key $sk =$ an integer $mod\, n$
    - Public key $pk$ is $sk\,.\,G$

# Signing a Transaction

- Start with a message `m`;
- Compute signature $\sigma = Sign(sk, m)$
- Broadcast $(tx, \sigma, pk)$
- Each node verifies: $Verify(pk, m, \sigma) = True$
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